

Problem Set 3

Probability and Statistical Inference

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Reference formulas for Question 1. As in Lecture 3, choosing a model matters; memorising the gallery formulas is non-examinable. You may use the following.

Distribution	Mean; variance	Useful probability
$N \sim \text{Poisson}(\lambda)$	$\lambda; \lambda$	$\Pr(N = j) = e^{-\lambda} \lambda^j / j!$
$W \sim \text{Exponential}(\lambda)$	$1/\lambda; 1/\lambda^2$	$\Pr(W > w) = e^{-\lambda w}, w \geq 0$
$G \sim \text{Gamma}(a, \lambda)$	$a/\lambda; a/\lambda^2$	shape-rate
$B \sim \text{Binomial}(m, p)$	$mp; mp(1-p)$	
$H \sim \text{Geometric}(p)$	$1/p; (1-p)/p^2$	$H = 1, 2, \dots$

1. **Choosing probability distributions.** Use the model for counts and waiting times from Lecture 3 for calls arriving at a support center at a constant rate of 3 per hour. Counts in disjoint time intervals are independent. Independently across calls, each call results in a sale with probability 0.2.
 - (a) Let N be the number of calls in one hour. State its distribution, find $\mathbb{P}(N = 2)$, and give its expectation and variance.
 - (b) Let W_1 be the waiting time, in hours, to the next call. State its distribution, find $\mathbb{P}(W_1 > 1/2)$, and give its expectation.
 - (c) Let W_4 be the waiting time to the fourth call. State its distribution using the shape-rate parameterization, and give its expectation and variance.
 - (d) Let S be the number of sales among the next ten calls, and let H be the number of calls up to and including the first sale. State both distributions. Find $\mathbb{E}[S]$, $\text{Var}(S)$, $\mathbb{P}(H > 4)$ and $\mathbb{E}[H]$.
 - (e) A researcher needs a distribution for an unknown proportion that can take values only between zero and one. The researcher also needs a distribution for a positive variable produced by repeated multiplicative changes (for example, repeated percentage growth). Which two distributions from the lecture are natural starting points? Briefly explain each choice.

2. **Normal vectors and related distributions.** Suppose

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N_2 \left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 4 & 2 \\ 2 & 9 \end{pmatrix} \right).$$

- (a) Give the marginal distributions of X_1 and X_2 , and calculate their correlation.
- (b) Find the distribution of $Y = 3 + X_1 - 2X_2$.
- (c) Find the value of a for which X_1 and $X_2 - aX_1$ are independent. Also give the distribution of $X_2 - aX_1$ at this value of a . Explain carefully which step of your argument uses joint normality.

- (d) Write $\mu = \mathbb{E}[X]$ and $\Sigma = \text{Var}(X)$. By multiplying out, verify that

$$(X - \mu)' \Sigma^{-1} (X - \mu) = \left(\frac{X_1 - 1}{2} \right)^2 + \left(\frac{X_2 - X_1/2 + 3/2}{\sqrt{8}} \right)^2.$$

Use part (c) and the lecture's definition of χ^2 to find its distribution.

- (e) Let $Z \sim N(0, 1)$ and $Q \sim \chi_\nu^2$ be independent, and define $T = Z/\sqrt{Q/\nu}$. Use the definitions of the chi-squared and F distributions to show that $T^2 \sim F_{1,\nu}$.

3. **Tail behavior and moments.** For $a > 0$, consider the following candidate probability density. Calling it a candidate means that you must still check that it is a valid density:

$$f_a(x) = \frac{a}{2(1 + |x|)^{a+1}}, \quad -\infty < x < \infty.$$

- (a) Verify that f_a is a probability density.
 - (b) For $r > 0$, determine exactly when $\mathbb{E}[|X|^r]$ is finite. It is sufficient to compare the integrand with a power of x in the tails; you do not need to evaluate the moment when it is finite.
 - (c) For which values of a does the mean exist? When it exists, what is it? Explain why symmetry alone is not enough to answer this question.
 - (d) For which values of a does the variance exist? Give one value of a for which the mean exists but the variance does not.
 - (e) For $M \geq 0$, show that $\mathbb{P}(|X| > M) = (1 + M)^{-a}$. Explain why a tail probability can approach zero even when some moments are infinite, so this result does not contradict your earlier answers.
4. **Using Jensen's inequality.** You may use Jensen's inequality for both convex and concave functions. For a concave function, the direction of the usual convex inequality is reversed. Let W be a positive random variable for which all expectations used below exist.
- (a) Determine which is larger, $\mathbb{E}[\log W]$ or $\log \mathbb{E}[W]$. State when the inequality is strict, and give an intuitive interpretation.
 - (b) Determine which is larger, $\mathbb{E}[1/W]$ or $1/\mathbb{E}[W]$.
 - (c) Let $0 < s < r$ and suppose $\mathbb{E}[|X|^r] < \infty$. By applying Jensen to a suitable function of $|X|^r$, prove that

$$\mathbb{E}[|X|^s] \leq (\mathbb{E}[|X|^r])^{s/r}.$$

Explain what this proves about the existence of lower-order moments.

- (d) Apply Jensen to a square to prove that $\text{Var}(X) \geq 0$. Under the usual finite-second-moment condition, when does equality hold?

5. **Regression matrices and error covariance.** For three observations, let z be the explanatory variable and y be the outcome variable, with

$$z = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \quad y = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix},$$

and include an intercept in the regression of y on z .

- (a) Write down the design matrix X and state the dimensions of y , X and β .
- (b) Calculate $X'X$ and $X'y$. Using the formula $\hat{\beta} = (X'X)^{-1}X'y$, calculate the OLS intercept and slope.
- (c) Calculate the residual vector $\hat{u} = y - X\hat{\beta}$ and verify $X'\hat{u} = 0$.
- (d) Suppose, conditional on X , the error covariance matrix is

$$\Omega = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{pmatrix}.$$

Use the diagonal and off-diagonal entries of Ω to describe its pattern of heteroscedasticity and dependence. Then calculate

$$\text{Var}(\hat{\beta} \mid X) = (X'X)^{-1}X'\Omega X(X'X)^{-1}.$$

- (e) If the errors instead had covariance I_3 , their contribution to $\text{Var}(\hat{\beta} \mid X)$ would be different. Would this change the OLS coefficient formula used in part (b), or only its sampling variance? Explain using the formulas above.